

# LEON MOTAUNG

## AI Engineer pursuing Accounting Degree

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## PROFESSIONAL PROFILE

Highly motivated AI/ML Engineer and Web Developer with over 5 years of experience building and deploying scalable business systems. Currently combining technical innovation with financial analysis through a Bachelor of Commerce (BCom) in Financial Accounting at the University of the Western Cape. Proven expertise in Natural Language Processing (NLP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and Cloud Architecture (AWS certified). Passionate about leveraging machine learning to solve complex enterprise problems, improve workflows, and bridge the gap between technology and business.

## SKILLS & EXPERTISE

**AI / Machine Learning:** Neural Networks, LLM Fine-Tuning, NLP, Retrieval-Augmented Generation (RAG), Semantic Search

**Cloud & DevOps:** AWS (Solutions Architect, Developer, Cloud Practitioner, SysOps Admin), DNS, Serverless Architecture

**Programming & Math:** Python, NumPy, Pandas, Java (Basics), Linear Algebra, Calculus, Statistics, Optimization

**Web Development:** HTML, CSS, Tailwind CSS, Bootstrap, JavaScript, React, Redux, Node.js, Express, PHP, Flask

**Databases & Tools:** MySQL, MongoDB, Vector DBs (Pinecone), Git, Linux, Windows, Excel, Adobe Photoshop

## PROFESSIONAL EXPERIENCE

### AI/ML & Software Engineer | DeWet Technologies (Pty) Ltd

Jan 2022 - Present

- Built and maintained scalable web applications and business systems, improving performance and deployment workflows.
- Designed responsive UI/UX using React, Tailwind CSS, and JavaScript, and integrated Node.js/Express, PHP, and MySQL backends.
- Implemented MongoDB-based data models, custom RESTful APIs, and cloud infrastructure on AWS.
- Developed Semantic Search systems and Retrieval-Augmented Generation (RAG) systems using IBM and AWS compute.
- Spearheaded the Tlou project (2026) - training an 800M Parameter LLM supporting 11 official South African languages.

### Founder & Private Tutor | Lyro Tutor Platform

2026 - Present

- Founded the platform to deliver personalized academic tutoring, incorporating custom curriculums and learning materials.
- Instruct Grade 10 learners in Mathematics and Mathematical Literacy, improving comprehension and test scores.
- Designed and built the tutoring platform with integrated Yoco payment gateway and scheduling systems.

## PROFESSIONAL EXPERIENCE (CONTINUED)

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### Web Developer & Guest Relations Manager | Sefatsa Guest House

*Apr 2026 - Jun 2026*

- Revamped the official website, introducing booking availability tracking and modern mobile-first UI.
- Managed shift procurement, guest reservations, agency invoicing, and customer care for late-night check-ins.
- Conducted brand awareness campaigns, managing inventory and marketing on Facebook and TikTok.

### Project Leader | Women in Computer Science (WICS)

*2025 (1 Year)*

- Led community and educational projects promoting gender diversity and inclusion in technology and computer science.
- Organized and mentored technical bootcamps, programming workshops, and technical hackathons.

### Computer Technician / Web Developer | SANCATDRA

*Jun 2021 - Dec 2021*

- Developed responsive web applications, applied SEO best practices, and managed domain migrations & DNS configurations.
- Provided on-site IT support, system troubleshooting, and image editing using Adobe Photoshop.

### Computer Technician | Mafube Local Municipality

*Nov 2019 - Feb 2020*

- Diagnosed and resolved hardware, software, operating system, and network switch/router issues.
- Maintained system backups, performed system upgrades, and ensured stable municipal network connectivity.

## EDUCATION & CERTIFICATIONS

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### BCom in Financial Accounting

*2022 - Present*

University of the Western Cape (UWC)

- FNB App Academy - Full Stack Development Certification (2025)
- AWS Certified Solutions Architect, Cloud Developer, SysOps Administrator (2024)
- Fundamentals in Generative AI (AWS Cloud Infrastructure & Inference) (2025)
- AWS Educate - Introduction to Generative AI Training (2025)
- AWS Fundamentals to Machine Learning Certification (2025)
- IBM Python Development for Data Science (2024)
- Udemy Complete Web Development Course (15 Projects) (2021)

## HACKATHONS & LEADERSHIP ACTIVITIES

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- **Stellar Blockchain Ideathon Judge (2025):** Evaluated Web3 projects on technical code execution, usability, and business viability.
- **Google G20 Hackathon (2025):** Co-developed an educational software prototype leveraging Retrieval-Augmented Generation (RAG) to ensure high-accuracy educational resources.
- **NASA Space Apps Hackathon (2025):** Completed the 'Exoplanet Hunting' challenge, developing algorithms to process and visualize stellar transit data for planetary detection.
- **Apart Research AI Safety Hackathons (2025):** Researched and prototyped LLM jailbreak defense scanners and model alignment assessment tools.
- **YES Program Energy Attendee (2025):** Selected for youth leadership training in sustainable energy systems hosted at the CTICC.
- **Telkom Hackathon & OceanHub Africa (2025):** Competed in rapid-prototyping tracks for digital accessibility and ocean monitoring technologies.